Sai Vikas Reddy Yeddulamala

AI/ML Engineer | **Location:** Raleigh, NC | **Mobile**: 919-672-0116 | **Email:** saivikas.y@zohomail.com

SUMMARY:

AI/ML Engineer with + years of experience building and deploying production AI systems across healthcare and banking, spanning agentic LLM workflows, RAG architectures, and classical ML at scale. Currently at Cardinal Health architecting LangGraph-based compliance agents, hybrid retrieval RAG systems, and LoRA-fine-tuned SLMs deployed via vLLM and Triton on AWS. Track record of pairing modern GenAI techniques with rigorous evaluation, A/B testing, and statistical analysis to deliver measurable business impact. AWS Certified ML Engineer and Azure Data Scientist Associate.

SKILLS:

Generative AI & LLM Systems: LangChain, LangGraph, MCP, Multi-Agent Orchestration, Tool Calling, RAG, GraphRAG, Hybrid Retrieval (BM25 + Dense), Rerankers (ColBERT, Cohere), Vector DBs (FAISS, Pinecone), LoRA/QLoRA, PEFT, DPO, GRPO, DSPy, Structured Outputs (Instructor, Pydantic), Guardrails, Hugging Face Transformers, vLLM, Triton, TensorRT-LLM, FlashAttention, KV-Cache, Quantization, ONNX
Machine Learning & Deep Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, CatBoost, CNNs, RNNs, LSTMs, Transformers, GraphSAGE, TCN, Two-Tower Models, DeepFM, Distributed Training (FSDP, DeepSpeed), SHAP
Statistics & Experimentation: A/B Testing, Power Analysis, Causal Inference (Difference-in-Differences, Propensity Score Matching), Uplift Modeling, Hypothesis Testing, Regression, Time Series Forecasting, Conformal Prediction, Bayesian Methods
MLOps & Evaluation: MLflow, Kubeflow, Docker, Kubernetes (EKS, AKS), CI/CD (CodePipeline, Azure DevOps, Jenkins), Model Registry, Drift Detection (Evidently), Shadow Deployments, Canary Releases, Feature Stores (Feast), LLM Evaluation (RAGAS, Langfuse, Promptfoo)
Data Engineering & Cloud: Apache Spark (PySpark), Apache Kafka, Apache Airflow, Snowflake, Databricks, Delta Lake, Cassandra, AWS (SageMaker, S3, EC2, EKS, Lambda), Azure (Azure ML, AKS, Synapse), GCP (Vertex AI)
Programming & Tools: Python, SQL, Bash, Java, Git, FastAPI, REST APIs, gRPC, Streamlit, Tableau, Power BI, Linux

EDUCATION:

Master's in Computer Science, Data Science specialization - North Carolina State University **Aug 2023 - May 2025**
Bachelor of Technology in Electronics and Communication Engineering - GITAM University, India **June 2017 - May 2021**

EXPERIENCE:

Cardinal Health, NC | Nov 2024 – Current | ML Engineer

- Built a production agentic compliance workflow on LangGraph with MCP tool integration, stateful routing, and human-in-the-loop guardrails, ingesting regulatory feeds and internal alerts to auto-triage recall and shortage events for same-day analyst response.
- Optimized a RAG-based research assistant with hybrid BM25 + dense retrieval, knowledge-graph entity lookup, ColBERT reranking, and a RAGAS eval harness for faithfulness and precision/recall, improving retrieval precision **3×** and cutting analyst research time **40%**.
- Fine-tuned 7B-parameter SLMs via LoRA/QLoRA for structured information extraction from clinical and regulatory documents (OCR + LLM), achieving 98% exact-match on critical fields with schema-validated JSON outputs and human-in-the-loop review.
- Contributed to MLOps tooling on AWS EKS using MLflow, Docker, CodePipeline (CI/CD), and Evidently drift monitoring, reducing model deployment time from weeks to days and catching prediction degradation pre-production.
- Optimized LLM inference serving with vLLM and NVIDIA Triton using dynamic batching and FlashAttention, improving throughput **8×** and bringing p95 latency under **150ms**, enabling near real-time scoring for downstream compliance and logistics applications.
- Developed a pharmaceutical demand forecasting model using LightGBM with temporal cross-validation and uncertainty estimation across **2,000+ SKUs**, reducing stockout incidents by **20%** and supporting inventory planning decisions across distribution centers.

Tata Consultancy Services, India | June 2022 – June 2023 | Data Scientist

- Built and deployed a customer-facing self-service banking assistant on AKS combining intent classification, hybrid BM25 + FAISS retrieval, and a fine-tuned transformer with confidence-based fallback to human agents, trained on **200K+** interactions, achieving **~55% containment**, **40%** per-contact cost reduction, and **12%** CSAT lift.
- Developed a fraud detection system combining GraphSAGE embeddings on the transaction-entity graph (card, device, merchant) with an XGBoost classifier and a TCN for short behavioral sequences, increasing precision at 95% recall by **4.5%** and cutting scoring latency **20%** on Azure Event Hubs and AKS.
- Designed A/B tests with proper power analysis and post-launch lift quantification using matched-cohort comparisons to attribute business impact of fraud and chatbot rollouts, providing statistically defensible estimates that informed go/no-go decisions for production deployment.

Tata Consultancy Services, India | June 2021 – May 2022 | Junior Data Scientist

- Built a two-stage recommendation system using two-tower neural retrieval for candidate generation with ANN search and a DeepFM model to re-rank top-K with eligibility and behavioral features, served in real time via GraphQL, delivering **7-9% lift in approved applications** validated through A/B testing.
- Developed churn and loan-default models using LightGBM and CatBoost with seasonality, trend, and holiday features in Snowflake, paired with SHAP explanations surfaced in Power BI for stakeholders, contributing to a **4-6% reduction in churn** and **3-4% fewer high-risk loan approvals**.

CERTIFICATIONS

AWS Certified Machine Learning Engineer – Associate (MLA-C01) • Microsoft Certified: Azure Data Scientist Associate (DP-100)